

Notes: Doraszelski, Lewis, and Pakes (AER, 2018)

Just Starting Out: Learning and Equilibrium in a New Market

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1 Big picture

- **Question.** How do firms learn to behave in a newly deregulated market, and when is it reasonable to replace learning dynamics with an equilibrium model?
- **Setting.** The paper studies the United Kingdom market for *frequency response* (FR), a service National Grid (NG) uses to stabilize the electricity system in real time. The market was deregulated in November 2005 under CAP047, replacing mandatory provision at an administered price with monthly bidding.
- **Core challenge.** Firms entered the new market with considerable **strategic uncertainty** about rivals' bids and **demand uncertainty** about how NG would allocate FR as a function of bids. The main empirical challenge is to separate learning dynamics from changes in fundamentals.
- **Main descriptive result.** The market evolves through three phases: an **early rising-price phase** with experimentation and a price bubble, a **middle falling-price phase** with strong competitive adjustment and convergence, and a **late stable-price phase** in which bids fluctuate in a narrow range around a rest point.
- **Main structural contribution.** The paper first estimates demand and marginal cost primitives under a **weak rationality** assumption that is much less demanding than assuming equilibrium. It then uses these primitives to compare a complete-information Nash equilibrium with learning models.
- **Main learning result.** During the middle phase, **fictitious play** models—especially those placing greater weight on recent rival bids—combined with **adaptive learning** about the demand elasticity fit the data better than a complete-information Nash equilibrium. In the late phase, the equilibrium model performs about as well as the best learning models.
- **Why this paper matters.** It shows that after a market design change, equilibrium can be a poor short-run description even if it eventually becomes a good long-run description. For counterfactual work, the transition path may need a learning model rather than immediate equilibrium.

2 Data and measurement (key objects)

2.1 Observed market environment

- **Institutional setting.** About 98% of UK electricity is sold via bilateral forward contracts, while about 2% is balanced in real time through the balancing mechanism (BM). BM units submit contracted positions one hour before settlement, and NG then rebalances supply and demand.
- **Frequency response as a separate product.** FR is distinct from electricity itself. NG must maintain a required amount of FR capacity at all times to keep system frequency within a 1% band around 50Hz.
- **Unit of exchange.** In the FR market, BM units submit a *monthly bid* in pounds per MWh of FR capacity. If NG calls on a unit, it receives a holding payment equal to its bid times the quantity of FR provided.

- **Why a unit may not be chosen even with a low bid.** NG’s dispatch is not a simple lowest-bid auction. Product differentiation arises from the precision of governor controls, the unit’s current operating position, and transmission constraints that may make a unit ineligible.

2.2 Observed panel and sample construction

- **Sample period.** November 2005 to October 2011, the first six years after deregulation.
- **Main data sources.** From National Grid: monthly bids and daily quantities of FR provided by each BM unit. From Elexon: half-hourly BM bids and acceptances plus contracted positions, which help recover operating position. Additional data on ownership, station characteristics, and fuel prices are collected from other sources.
- **Market participants.** There are 130 BM units grouped into 61 power stations owned by 29 firms. The market is mildly concentrated, with a 10-firm concentration ratio of 84% and an HHI of 0.11.
- **Estimation sample for demand and learning.** The analysis focuses on the 72 BM units owned by the 10 largest firms, which together account for just over 80% of revenue in the FR market. The remaining BM units are bundled into an outside good.
- **Large firms.** Drax alone accounts for 23.8% of revenue over the sample period, followed by E.ON at 16.0% and RWE at 11.6%.

2.3 Key measured objects

- **Market share.** $s_{j,t}$ is BM unit j ’s share of FR in month t .
- **Eligibility.** $e_{j,t} = 1(s_{j,t} > 0)$ indicates whether BM unit j receives positive share in month t .
- **Bid.** $b_{j,t}$ is the monthly bid tendered by BM unit j for month t .
- **Observable demand shifters.** $x_{j,t}$ includes: the fraction of the month the unit is fully loaded, the fraction part-loaded, a missing-operating-position indicator, and whether the unit is under contract for positive firm frequency response (FFR) volume.
- **Descriptive statistics.** In the estimation sample, mean market share is 0.011, eligibility is 0.752, and bids average £5.453/MWh with a standard deviation of 2.759.
- **Market size.** M_t is the total FR requirement in month t . This is regulated and exogenous to bids.

2.4 Unobservables and timing

- **Unobserved demand component.** $\xi_{j,t}$ is a BM-unit-month unobservable that shifts NG’s demand for unit j .
- **Persistent unobservables.** The paper allows

$$\xi_{j,t} = \rho \xi_{j,t-1} + \nu_{j,t},$$

where $\nu_{j,t}$ is i.i.d. and mean independent of current and past bids and observables.

- **Timing.** Bids for month t are submitted before the twentieth of month $t - 1$. By then, firms observe rivals' bids up through the current month and realized quantities up through the previous month.
- **Economic interpretation.** This timing is what makes learning feasible: firms repeatedly observe rivals' actions and the realized allocation rule, then update future bids.

3 Models and empirical specifications (all equations + interpretation)

3.1 Demand for frequency response

The paper approximates NG's allocation rule with a generously parameterized logit model at the BM-unit-month level:

$$s_{j,t} = \frac{e_{j,t} \exp(\alpha \ln b_{j,t} + \beta x_{j,t} + \gamma_j + \mu_t + \xi_{j,t})}{1 + \sum_k e_{k,t} \exp(\alpha \ln b_{k,t} + \beta x_{k,t} + \gamma_k + \mu_t + \xi_{k,t})}.$$

- γ_j is a BM-unit fixed effect capturing time-invariant desirability, such as governor precision.
- μ_t is a month fixed effect capturing shifts in the outside good and common demand conditions.
- The coefficient α is the key price-sensitivity parameter.

Interpretation. The logit model is not meant to be the literal dispatch rule used by NG's proprietary linear program. It is an empirical approximation that provides tractable own- and cross-price elasticities in a market with many units.

3.2 Eligibility equation

Because many BM units receive zero market share, the paper pairs the logit demand system with a probit eligibility model:

$$e_{j,t} = 1 \left(\tilde{\beta} x_{j,t} + \tilde{\gamma}_j + \tilde{\mu}_t + \eta_{j,t} > 0 \right),$$

so that

$$\Pr(e_{j,t} = 1 \mid x_{j,t}) = \Phi \left(\tilde{\beta} x_{j,t} + \tilde{\gamma}_j + \tilde{\mu}_t \right).$$

Interpretation. A unit may be absent because of maintenance, transmission constraints, or very high bids. The eligibility equation captures whether it is even in the running for FR in a given month.

3.3 Linearized logit and the proxy for persistent unobservables

Conditional on positive share, the logit equation can be written as

$$\delta_{j,t} \equiv \ln s_{j,t} - \ln s_{0,t} = \alpha \ln b_{j,t} + \beta x_{j,t} + \gamma_j + \mu_t + \xi_{j,t}.$$

If $\rho \neq 0$, then $\xi_{j,t}$ is persistent and correlated with bids chosen after observing $\xi_{j,t-1}$. To correct for this simultaneity, the paper quasi-first-differences the equation:

$$\delta_{j,t} - \rho\delta_{j,t-1} = \alpha(\ln b_{j,t} - \rho \ln b_{j,t-1}) + \beta(x_{j,t} - \rho x_{j,t-1}) + \tilde{\gamma}_j + \tilde{\mu}_t + \nu_{j,t}.$$

Interpretation. This is the paper's main demand-estimation step. It uses the timing of bid choices and the persistence of demand shocks to get around the correlation between current bids and current unobservables.

3.4 Demand estimates and economic meaning

The preferred NLLS estimates imply:

- $\hat{\alpha} \approx -1.614$,
- a positive and significant effect of being fully loaded,
- an even larger positive effect of being part-loaded,
- a negative effect of already having positive FFR volume,
- persistence in the demand unobservable with $\hat{\rho} \approx 0.406$.

Interpretation.

- Since market shares are small, α is close to the demand elasticity with respect to bids.
- Part-loaded units are especially attractive because NG prefers units in the middle of their operating range.
- Units already committed to FFR are less useful in the MFR market.

3.5 Profit and the weak-rationality approach to cost estimation

The realized profit of firm i in month t is

$$\pi_{i,t} = \sum_{j \in \mathcal{J}_i} (b_{j,t} - c_j) M_t s_j(b_t, x_t, \xi_t, e_t; \theta_0),$$

where c_j is the constant marginal cost of BM unit j .

The firm chooses bids to maximize perceived expected current profit,

$$\max_{b_{i,t}} \mathbb{E}_{b_{-i,t}, \xi_t, e_t, \theta_t} \left[\sum_{j \in \mathcal{J}_i} (b_{j,t} - c_j) M_t s_j(b_t, x_t, \xi_t, e_t; \theta_t) \mid \Omega_{i,t-1} \right].$$

This yields first-order conditions of the form

$$\mathbb{E} \left[M_t s_k(\cdot) + \sum_{j \in \mathcal{J}_i} (b_{j,t} - c_j) M_t \frac{\partial s_j(\cdot)}{\partial b_{k,t}} \mid \Omega_{i,t-1} \right] = 0, \quad \forall k \in \mathcal{J}_i.$$

Because the paper does not specify firms' full subjective belief process at this stage, it cannot estimate cost from those pointwise FOCs directly. Instead, it imposes a **weak rationality** condition: firms do not make *systematic* mistakes in their perceived FOCs during the late phase. Therefore the time average of the perceived FOCs is zero:

$$\frac{1}{T} \sum_{t=44}^{72} \left[M_t s_{k,t} + \sum_{j \in \mathcal{J}_i} (b_{j,t} - c_j) M_t (1\{k = j\} - s_{k,t}) \hat{\alpha} \frac{s_{j,t}}{b_{k,t}} \right] = 0, \quad \forall k \in \mathcal{J}_i.$$

Interpretation. This is a clever identification move. The paper does *not* need the market to already be in equilibrium. It only needs firms in the late phase to be, on average, “doing their best” and not repeatedly misperceiving profitable directions of adjustment.

3.6 Cost estimates and rationality diagnostics

The implied average marginal cost across the 72 BM units owned by the ten largest firms is about £1.40/MWh, with standard deviation 0.66 across units and average standard error 0.04.

Interpretation.

- These costs are plausible relative to the pre-CAP047 administered “cost-reflective” price of about £1.7/MWh.
- Most variation in cost is across stations, not within a station.
- Residuals from the estimated FOCs are large and persistent in the early and middle phases, but much smaller in the late phase, which supports the paper’s claim that the late phase is the right place to estimate costs.

3.7 Complete-information Nash equilibrium

As a benchmark, the paper computes a complete-information Nash equilibrium (CINE). The bid vector b_t^* solves, for every BM unit k ,

$$\mathbb{E}_{\xi_t, e_t} \left[M_t s_k(b_t^*, x_t, \xi_t, e_t; \theta_0) + \sum_{j \in \mathcal{J}_i} (b_{j,t}^* - c_j) M_t \frac{\partial s_j(b_t^*, x_t, \xi_t, e_t; \theta_0)}{\partial b_{k,t}} \mid M_t, x_t, \xi_{t-1}, \theta_0, c \right] = 0.$$

Interpretation. Here firms know the true demand system and use the objective distribution of shocks. The only uncertainty left is about contemporaneous realizations of demand and eligibility.

3.8 Fictitious play for strategic uncertainty

To model uncertainty about rivals’ bids, the paper assumes firms form beliefs from the empirical distribution of past rival play. Rivals’ bid vectors from month $\tau \leq t - 1$ receive weight proportional to

$$\delta^{t-\tau-1}, \quad \delta \in [0, 1].$$

- $F(0)$: full decay beyond the immediate past; the firm best responds to last month’s bids with certainty.

- $F(1)$: equal weights on all past observations.
- Intermediate δ values give more weight to recent history without ignoring older observations.

Interpretation. This is the paper’s model of *strategic uncertainty*. It lets firms use observed bidding history to form beliefs about what competitors will do next.

3.9 Adaptive learning for demand uncertainty

To model uncertainty about demand parameters, the paper lets firms sequentially re-estimate the demand system using the data available at the time they bid. In month $t - 1$, firms only observe data up to month $t - 2$ for realized quantities.

The paper considers five cases:

- $A(\alpha)$: firms learn only about the bid sensitivity parameter.
- $A(\alpha, \beta)$: firms learn about bid sensitivity and the coefficients on time-varying characteristics.
- $A(\alpha, \mu)$: firms learn about bid sensitivity and month effects.
- $A(\alpha, \beta, \mu)$: firms learn about all of the above.
- $A(\emptyset)$: no demand uncertainty; firms use full-sample demand estimates.

Interpretation. This is the paper’s model of *structural uncertainty*. The main question is whether firms are still learning how price-sensitive NG is and how the outside option evolves over time.

3.10 Predicted bids under learning

Given a fictitious play model $F(\delta)$, an adaptive learning model $A(y)$, and the information set D_t , predicted bids solve expected-profit FOCs:

$$\mathbb{E}_{b_{-i,t}, \xi_t, e_t, \theta_t} \left[M_t s_k(b_{i,t}^{(\delta,y)}, b_{-i,t}, x_t, \xi_t, e_t; \theta_t) + \sum_{j \in \mathcal{J}_i} (b_{j,t}^{(\delta,y)} - c_j) M_t \frac{\partial s_j(b_{i,t}^{(\delta,y)}, b_{-i,t}, x_t, \xi_t, e_t; \theta_t)}{\partial b_{k,t}} \mid F(\delta), A(y), D_t \right] = 0.$$

The paper evaluates these predictions in two ways:

- **Single-period predictions:** each month uses actual past data.
- **Multi-period predictions:** simulated bids and shares overwrite the data as the model is iterated forward.

Interpretation. Single-period predictions ask whether the model gets the next move right. Multi-period predictions ask whether the model generates a realistic transition path.

3.11 Measures of fit

The first fit measure is the average absolute prediction error in NG's FR price:

$$\text{CE}(\delta, y) = \frac{1}{T} \sum_t \left| \sum_j b_{j,t}^{(\delta, y)} \frac{s_{j,t}}{\sum_j s_{j,t}} - \sum_j b_{j,t} \frac{s_{j,t}}{\sum_j s_{j,t}} \right|.$$

The second is the share-weighted mean square prediction error at the BM-unit level:

$$\text{MSE}(\delta, y) = \frac{\sum_t \sum_j (b_{j,t}^{(\delta, y)} - b_{j,t})^2 s_{j,t}}{\sum_t \sum_j s_{j,t}}.$$

Interpretation. The first tells us how well the model predicts the price NG pays. The second tells us how well it predicts individual-unit bids.

3.12 Efficiency calculation

To ask whether out-of-equilibrium behavior had major welfare or efficiency consequences, the paper compares the average cost per MWh under actual shares to the average cost under the equilibrium shares. The actual average cost is £1.318/MWh, versus £1.3133/MWh in the counterfactual equilibrium.

Interpretation. The transition dynamics mattered a lot for prices and learning, but not much for productive efficiency. The main disequilibrium effect is redistribution from NG to firms rather than large productive waste.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no external IV; identification comes from design and timing

- This paper does not rely on a traditional excluded instrument.
- Identification instead comes from: the market's institutional timing, the relative stability of fundamentals through much of the sample, and distinct behavioral restrictions for demand, cost, and learning.

4.2 Why the descriptive phase decomposition is credible

- The demand and supply environment is relatively stable for roughly the first three-and-a-half years.
- MFR quantity, seasonality, participation, and fuel prices do not move in ways that explain the large early and middle-phase bid swings.
- This makes it plausible to read the price bubble and subsequent decline as learning dynamics rather than just fundamentals.

4.3 Why demand is identified

- Bids vary across units and time, while operating-position variables and FFR exposure shift demand independently of bids.
- BM-unit fixed effects absorb time-invariant unit desirability.
- Month fixed effects absorb aggregate shocks and outside-good variation.
- The AR(1) structure for $\xi_{j,t}$, plus the timing of bid submission, allows the quasi-differenced NLLS equation to handle simultaneity between bids and persistent unobserved demand shocks.

4.4 Why cost is identified without assuming equilibrium

- A naive equilibrium-based approach would assume the market is already at Nash equilibrium in the late phase.
- The paper instead uses the weaker assumption that firms do not make systematic FOC mistakes on average in the late phase.
- This is attractive because it permits estimation even if the market has only converged to a *rest point* rather than a literally exact equilibrium.

4.5 Why the learning-versus-equilibrium comparison is informative

- The complete-information Nash equilibrium fixes beliefs at the objective model and lets only fundamentals move.
- The learning models allow beliefs about rivals and/or demand to evolve over time.
- Therefore if learning models systematically fit the middle phase better, the gap can be interpreted as evidence that firms had not yet converged to equilibrium behavior.

4.6 Robustness and remaining threats

- **Fuel prices.** Allowing marginal cost to vary with fuel prices changes little; the measured effect is economically small and hard to distinguish from a time trend.
- **Repositioning incentives in the BM.** Adding profits from BM repositioning does not materially change marginal cost estimates; the implied markup is small and insignificant.
- **Collusion.** Several exercises—correlations in bid changes across firms, cost recovery under collusive conduct, and bounds on rival-profit weights—provide little evidence of collusion.
- **Switching costs.** The timing of bid updates is not strongly explained by ex post lost profit, which weakens standard switching-cost stories as the main explanation for infrequent adjustment.

4.7 Relation to the literature

- On the theory side, the paper connects to belief-based learning in games, especially fictitious play.

- On the macro side, it borrows adaptive learning from Sargent and Evans-Honkapohja.
- On the empirical IO side, it shows how to estimate cost under weak rationality and how to compare equilibrium and learning models in a real market rather than a laboratory setting.

5 Tables and figures

5.1 Figure 1 and Figure 2: the market institution and the impact of deregulation

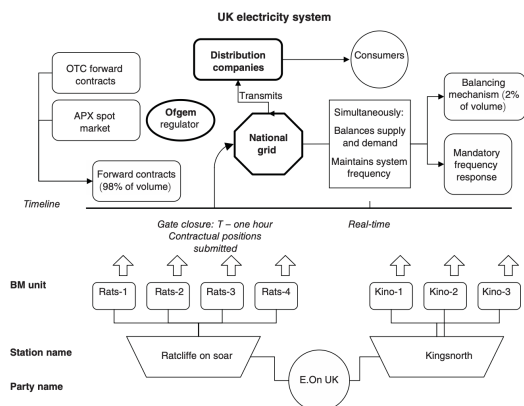


FIGURE 1. OVERVIEW OF THE UK ELECTRICITY MARKET

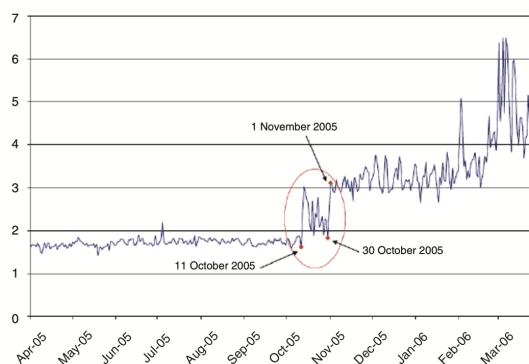


FIGURE 2. HOLDING PAYMENT IN £/MWh FOR HIGH RESPONSE BY DAY PRE AND POST CAP047

Read:

- Figure 1 is the institutional map: most electricity is sold through forward contracts, but NG balances the final 2% through the BM while simultaneously procuring FR.
- Figure 2 is the event-study style picture around CAP047. The holding payment jumps sharply after deregulation.
- This is the first motivating fact of the paper: marketization changed pricing immediately, not just gradually.
- It also motivates NG's original concern that a competitive market for FR could still produce high prices because demand is regulated and inelastic.

5.2 Table 1 and Figures 3–5: participants and why fundamentals are not the whole story

Read:

- Table 1 shows that the FR market is economically meaningful. Drax earns roughly £99.4 million over the sample, or about £1.4 million per month.
- Figure 3 shows that MFR quantity is volatile and seasonal, but not increasingly volatile early in the sample when the biggest bid changes occur.
- Figure 4 shows that FFR is fairly stable until July 2010, when it nearly doubles; fuel prices trend up mostly late in the sample.

TABLE 1—LARGEST FIRMS

Rank	Firm name	BM units owned	Total revenue	Revenue share (%)	Cumulative share (%)
1	Drax Power Ltd	6	99.4	23.8	23.8
2	E.ON UK plc	20	67.0	16.0	39.9
3	RWE plc	23	48.4	11.6	51.6
4	Eggborough Power Ltd	4	29.8	7.1	58.7
5	Keadby Generation Ltd	9	24.2	5.8	64.5
6	Barking Power Ltd	2	17.8	4.2	68.8
7	SSE Generation Ltd	4	15.2	3.6	72.5
8	Jade Power Generation Ltd	4	15.0	3.6	76.1
9	Centrica plc	8	14.7	3.5	79.6
10	Seabank Power Ltd	2	14.0	3.3	83.0

Notes: Firms ranked by revenue in the FR market. BM units owned is the maximum number owned over the sample period. Revenue is in inflation-adjusted millions of British pounds (base period is October 2011).

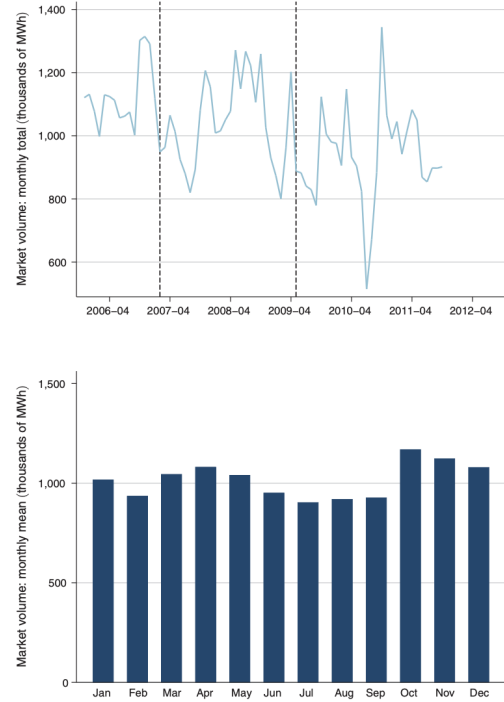


FIGURE 3. MFR QUANTITY BY MONTH (Top Panel) AND ON AVERAGE BY MONTH-OF-YEAR (Bottom Panel)

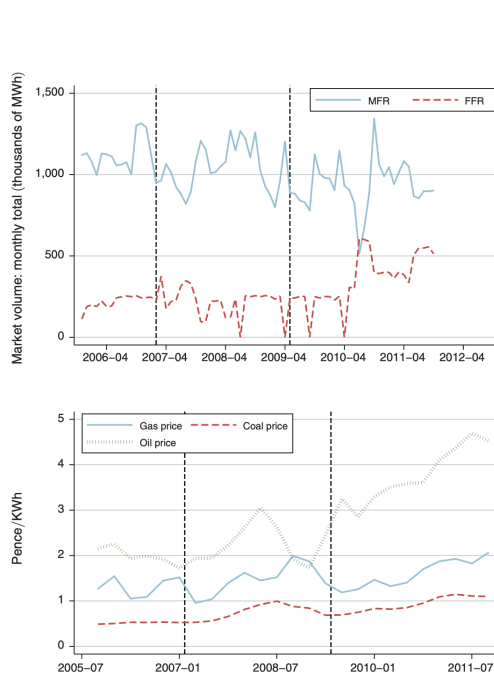


FIGURE 4. MFR AND FFR QUANTITIES BY MONTH (Top Panel) and FUEL PRICES BY QUARTER (Bottom Panel)

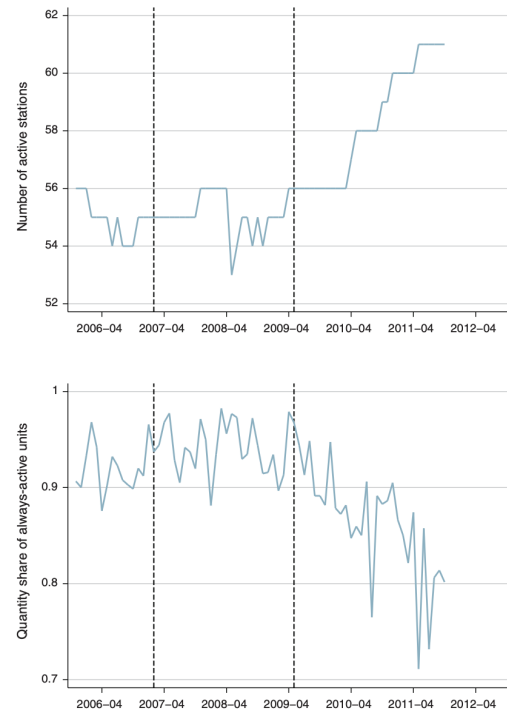


FIGURE 5. NUMBER OF ACTIVE POWER STATIONS BY MONTH (Top Panel) and MARKET SHARE OF ALWAYS-ACTIVE POWER STATIONS (Bottom Panel)

- Figure 5 shows that the number of active stations and the share of always-active units are fairly stable early on.
- These figures are crucial because they support the paper’s central identification claim: before mid-2009, most of the dramatic bid movement is hard to attribute to changing demand or supply conditions.

5.3 Figures 6–11: the three phases of learning and convergence

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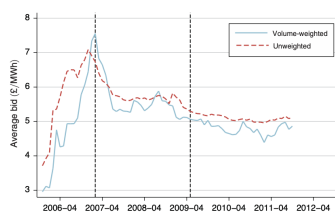


FIGURE 6. QUANTITY-WEIGHTED AND UNWEIGHTED FR PRICE BY MONTH

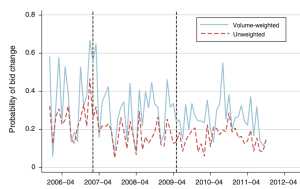


FIGURE 7. QUANTITY-WEIGHTED AND UNWEIGHTED PROBABILITY OF A BID CHANGE BETWEEN MONTH t AND $t - 1$

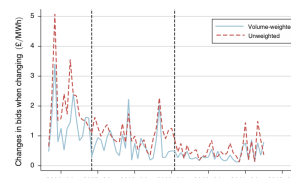


FIGURE 8. QUANTITY-WEIGHTED AND UNWEIGHTED ABSOLUTE VALUE OF BID CHANGE CONDITIONAL ON CHANGING BETWEEN MONTH t AND $t - 1$

Note: Weights are based on month $t - 1$ and are zero if the BM unit's bid did not change.

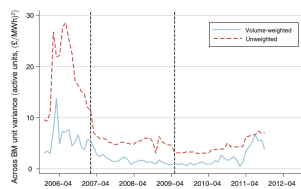


FIGURE 9. QUANTITY-WEIGHTED AND UNWEIGHTED VARIANCE IN BIDS ACROSS BM UNITS BY MONTH

Note: Weights are based on month t .

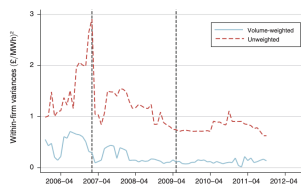


FIGURE 10. QUANTITY-WEIGHTED AND UNWEIGHTED VARIANCE IN BIDS BY MONTH WITHIN A FIRM (Top Panel)



FIGURE 10. QUANTITY-WEIGHTED AND UNWEIGHTED VARIANCE IN BIDS BY MONTH WITHIN A FIRM (Top Panel) AND ACROSS FIRMS (Bottom Panel)

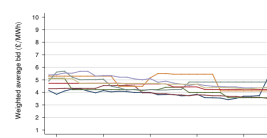
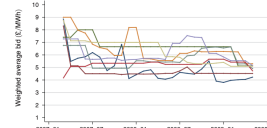
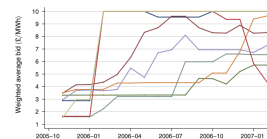


FIGURE 11. QUANTITY-WEIGHTED AVERAGE BIDS OF THE LARGEST POWER STATIONS BY MONTH

Note: November 2005–February 2007 (top panel), March 2007–May 2009 (middle panel), and June 2009–October 2011 (bottom panel). Stations ranked by revenue in the FR market during early and middle phases. Bids are censored above at 115 €/MWh to improve visual presentation.

Read:

- Figure 6 shows the three phases directly in the quantity-weighted FR price: rising early phase, falling middle phase, and stable late phase.
- Figures 7 and 8 show that bid changes are more frequent and much larger in the early phase, then smaller and less frequent later.
- Figures 9 and 10 show that cross-unit and especially across-firm bid dispersion collapses over time.
- Figure 11 is the richest descriptive figure. It shows that stations move very differently early on, converge aggressively in the middle phase, and then cluster tightly in the late phase.
- The economic story is that experimentation and heterogeneous beliefs dominate early, strong competitive undercutting dominates the middle, and a rest point emerges by the late phase.

5.4 Table 2, Table 3, and Figure 12: demand estimation works well

1*3

	Mean	Standard deviation	Min.	Max.
Market share	0.011	0.016	0.000	0.131
Eligibility	0.752	0.432	0.000	1.000
Bid	5.453	2.789	1.515	21.003
Fully loaded	0.133	0.236	0.000	0.997
Part-loaded	0.551	0.373	0.000	1.000
Missing operating position	0.115	0.319	0.000	1.000
Positive FFR volume	0.007	0.085	0.000	1.000

Notes: The unit of observation is a BM unit-month. The sample is restricted to BM units owned by the largest firms in Table 1.

	OLS	NLLS	ML
log bid	-1.648 (0.132)	-1.614 (0.119)	
Fully loaded	1.666 (0.220)	1.949 (0.182)	2.501 (0.355)
Part-loaded	2.111 (0.156)	2.234 (0.139)	2.168 (0.335)
Positive FFR volume	-0.794 (0.200)	-0.587 (0.245)	-0.500 (0.461)
Autocorrelation coefficient		0.406 (0.031)	
R^2 in shares	0.49	0.67	
Observations	3,831	3,509	5,175

Notes: OLS and NLLS estimates of logit model for market share and ML estimates of probit model for eligibility. All models include BM unit and month fixed effects. NLLS estimates allow the unobservable characteristic ξ_{it} to follow an AR(1) process with autocorrelation coefficient ρ . The unit of observation is a BM unit-month. Standard errors are clustered by BM unit. R^2 is for the fit of predicted versus actual market share, omitting observations with zero share.

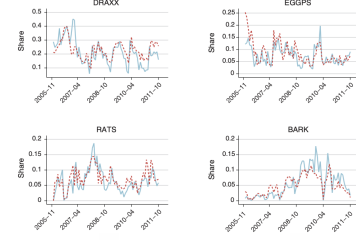


FIGURE 12. GOODNESS OF FIT

Notes: Realized (solid) and predicted (dashed) market share by month for the four largest power stations DRA00 (top left panel), EGGPS (top right panel), RATS (bottom left panel), and BARK (bottom right panel).

Read:

- Table 2 gives the basic scale of the data: mean market share 1.1%, eligibility 75.2%, and mean bid £5.453/MWh.
- Table 3 shows the demand estimates: $\hat{\alpha} \approx -1.61$ in the preferred NLLS specification, with strong positive effects of being fully loaded or part-loaded and a negative effect of positive FFR volume.
- The estimated persistence of the unobserved demand component is $\hat{\rho} \approx 0.406$.
- Figure 12 shows the fitted shares track the realized shares for large stations quite well.
- This is important because the whole structural analysis of learning later depends on having a credible reduced-form approximation to NG's allocation rule.

5.5 Tables 4–6 and Figure 13: plausible costs and convergence in residual mistakes

Read:

- Table 4 shows marginal costs between about £1.04 and £1.59/MWh across the largest stations. These are close to the pre-CAP047 administered benchmark of roughly £1.7/MWh.
- Table 5 shows that dual-fuel and oil units tend to have lower estimated cost than combined-cycle gas turbines, and later-vintage units may also be somewhat cheaper.
- Figure 13 shows the average residual from the estimated FOCs starts high in the early phase, turns negative in the middle phase, and becomes much smaller and trendless in the late phase.
- Table 6 shows residual autocorrelation declines over time. With all observations, the lagged residual coefficient falls from 0.542 in the early phase to 0.445 in the late phase; when restricting to bid changes, it falls all the way to 0.029 in the late phase.
- The message is that firms gradually stop making persistent directional mistakes when they actually revise bids.

TABLE 4—MARGINAL COST ESTIMATES FOR THE LARGEST POWER STATIONS

Station	BM units	Fuel	Vintage	Mean	Standard deviation
Barking	2	CCGT	1994	1.20	0.01
Connah's Quay	4	CCGT	1996	1.04	0.03
Cottam	4	Coal	1969	1.35	0.04
Drax	6	Coal	1974	1.06	0.04
Eggborough	4	Coal	1968	1.53	0.06
Peterhead	1	CCGT	2000	1.54	0.00
Ratcliffe	4	Coal	1968	1.33	0.06
Seabank	2	CCGT	1998	1.59	0.01

Notes: Characteristics of power station and average and standard deviation of marginal cost estimate c_j across BM units within station. Stations ranked by revenue in the FR market during early and middle phases.

TABLE 6—AUTOCORRELATION IN RESIDUALS

	All			Only bid changes		
	Early	Middle	Late	Early	Middle	Late
Lagged residual	0.542 (0.087)	0.343 (0.050)	0.445 (0.063)	0.389 (0.059)	0.126 (0.042)	0.029 (0.080)
R^2	0.63	0.48	0.20	0.74	0.38	0.08
Observations	1,080	1,931	2,088	355	449	401

Notes: Separate regressions of residual on lagged residual for the three phases of the FR market. All regressions include BM unit fixed effects. The unit of observation is a BM unit-month. The first three regressions use all observations. The last three regressions are restricted to observations in which a BM unit's bid changed between months. Standard errors are clustered by BM unit.

TABLE 5—MARGINAL COST ESTIMATES AND BM UNITS CHARACTERISTICS

	Coefficient
Unit vintage	−0.015 (0.017)
Dual fuel	−0.819 (0.466)
Large coal	−0.463 (0.429)
Medium coal	−0.683 (0.544)
Oil	−0.967 (0.397)
R^2	0.13
Observations	71

Notes: Regression of marginal cost estimate c_j on characteristics of BM unit. The omitted fuel type is combined cycle gas turbines (CCGT). The unit of observation is a BM unit. One observation is dropped because of missing vintage.

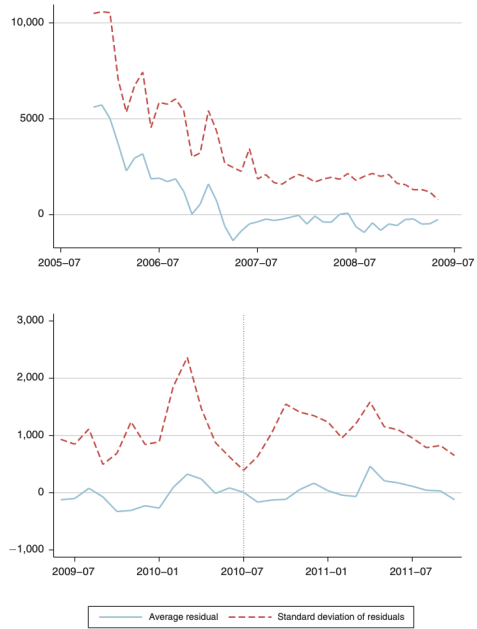


FIGURE 13. AVERAGE AND STANDARD DEVIATION OF RESIDUALS DURING THE EARLY AND MIDDLE PHASES (Top Panel) AND LATE PHASE (Bottom Panel)

Note: In the bottom panel, the dotted line indicates July 2010, when FFR volume nearly doubles.

TABLE 7—TIMING OF BID CHANGES AND EX POST PROFITABILITY

Rank	Firm name	Months changed	Matched direction (%)	Total profit	Ex post lost profit (%)	Ex post bid diff.	Ex post bid diff. (%)
1	Drax Power Ltd	23	80	68	2.0	1.1	20
2	E.ON UK plc	52	67	44	2.0	0.8	14
3	RWE plc	15	86	25	3.1	0.9	23
4	Eggborough Power Ltd	18	58	18	1.4	0.8	17
5	Keadby Generation Ltd	17	80	15	5.1	1.0	19
6	Barking Power Ltd	57	49	12	6.2	0.7	15
7	Jade Power Generation Ltd	15	54	10	4.7	1.2	24
8	SSE Generation Ltd	17	62	9	0.5	0.4	8
9	Seabank Power Ltd	9	89	9	8.2	1.1	20
10	Centrica plc	42	73	9	2.6	0.9	18

Notes: The unit of observation is a firm-month. The sample is restricted to the largest firms in Table 1. Profit is in millions of British pounds.

5.6 Table 7: heterogeneity in attention, updating, and foregone profits

Read:

- Some firms revise constantly, others rarely. Barking changes bids in 57 months and E.ON in 52, while Seabank changes in only 9 and RWE in 15.
- Frequent adjustment is not strongly related to total profit. The most active firms are not the most profitable firms.
- Ex post lost profits are usually modest: for example, Drax leaves about 2.0% of profit on the table, and RWE about 3.1%.
- But ex post bid differences are economically noticeable, often around 15%–24% away from the ex post optimal bid.
- This is an important clue: firms can be fairly far from ex post optimal bidding without losing huge amounts of money, which helps explain why convergence can take years.

5.7 Figure 14, Tables 8–10, and Figure 15: Learning beats equilibrium in the middle phase

Read:

- Figure 14 shows the sequential estimate of the demand elasticity parameter α starts too small in absolute value and gradually converges toward the full-sample estimate. That is direct evidence consistent with learning.
- Table 8 shows that in the middle phase the best average absolute prediction errors come from fictitious play with recent weighting and either no demand uncertainty or uncertainty only about α . The best cells are around 0.27–0.28, versus 0.53 for equilibrium under $A(\emptyset)$.
- Table 9 tells the same story in mean square error: values near 1.22 under the best learning models versus 1.45 for the equilibrium benchmark.
- Table 10 shows these differences are statistically and economically significant; the learning model beats equilibrium by about 0.19 in average absolute prediction error and by about 0.19 in weighted squared error on average.
- Figure 15 makes the mechanism visual: the equilibrium price path approaches the rest point too quickly, while the learning models mimic the gradual descent seen in the data.

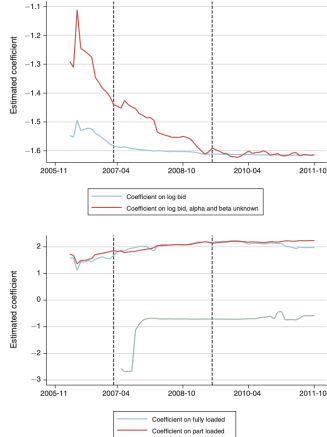


FIGURE 14. SEQUENTIAL ESTIMATES $\hat{\alpha}^{(t)}$ UNDER $A(\alpha)$ AND $A(\alpha, \beta)$ (Top Panel) AND $\hat{\beta}^{(t)}$ UNDER $A(\alpha, \beta)$ (Bottom Panel) BY MONTH

TABLE 8—MIDDLE PHASE: AVERAGE ABSOLUTE PREDICTION ERROR $CE^{(A, \alpha)}$

	Single-period prediction				Multi-period prediction	
	$A(\theta)$	$A(\alpha)$	$A(\alpha, \beta)$	$A(\alpha, \beta, p)$	$A(\theta)$	$A(\alpha)$
$F(0)$	0.33	0.28	1.29	0.50	0.93	0.31
$F(0.5)$	0.33	0.27	1.30	0.51	0.94	0.48
$F(1)$	0.51	0.44	1.03	0.36	0.67	0.28
Eq.	0.53	0.42	1.87	0.63	1.29	0.39

Note: Computed from bids predicted by fictitious play model $F(\theta)$ or complete information Nash equilibrium (rows) in combination with adaptive learning model $A(\alpha)$ (columns).



FIGURE 9. QUANTITY-WEIGHTED AND UNWEIGHTED VARIANCE IN BIDS ACROSS BM UNITS BY MONTH
Note: Weights are based in month t .

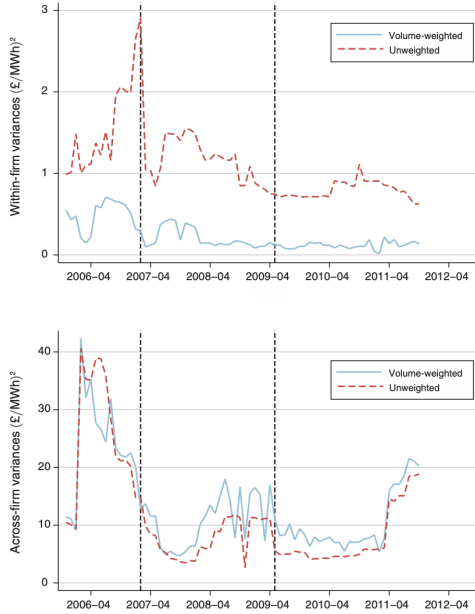


FIGURE 10. QUANTITY-WEIGHTED AND UNWEIGHTED VARIANCE IN BIDS BY MONTH WITHIN A FIRM (Top Panel) AND ACROSS FIRMS (Bottom Panel)

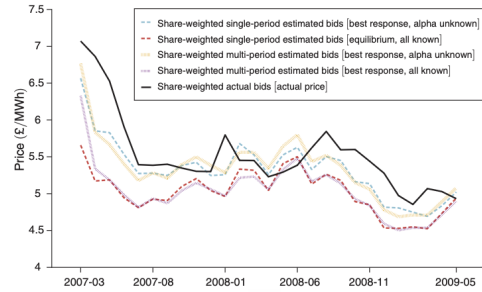


FIGURE 15. PREDICTED AND ACTUAL FR PRICE BY MONTH

Notes: Share-weighted average computed from bids predicted by fictitious play $F(\theta)$ with adaptive learning $A(\alpha)$ in both single and multi-period simulations, complete information Nash equilibrium without demand uncertainty $A(\theta)$, and actual bids (middle phase).

5.8 Tables 11–13 and Figure 16: by the late phase, equilibrium is nearly enough

TABLE 11—LATE PHASE: AVERAGE ABSOLUTE PREDICTION ERROR $CE^{(\delta,y)}$

	Single-period prediction					Multi-period prediction	
	$A(\emptyset)$	$A(\alpha)$	$A(\alpha, \mu)$	$A(\alpha, \beta)$	$A(\alpha, \beta, \mu)$	$A(\emptyset)$	$A(\alpha)$
$F(0)$	0.15	0.15	0.23	0.18	0.17	0.15	0.27
$F(0.5)$	0.15	0.15	0.24	0.19	0.17	0.15	0.27
$F(1)$	0.24	0.24	0.33	0.27	0.25	0.17	0.28
Eq.	0.17	0.17	0.28	0.21	0.19		

Note: Computed from bids predicted by fictitious play model $F(\delta)$ or complete information Nash equilibrium (rows) in combination with adaptive learning model $A(y)$ (columns).

TABLE 12—LATE PHASE: MEAN SQUARE PREDICTION ERROR $MSE^{(\delta,y)}$

	Single-period prediction					Multi-period prediction	
	$A(\emptyset)$	$A(\alpha)$	$A(\alpha, \mu)$	$A(\alpha, \beta)$	$A(\alpha, \beta, \mu)$	$A(\emptyset)$	$A(\alpha)$
$F(0)$	0.31	0.31	0.40	0.37	0.39	0.30	0.38
$F(0.5)$	0.32	0.32	0.40	0.38	0.39	0.30	0.38
$F(1)$	0.39	0.39	0.52	0.47	0.50	0.32	0.41
Eq.	0.31	0.31	0.44	0.38	0.38		

Note: Computed from bids predicted by fictitious play model $F(\delta)$ or complete information Nash equilibrium (rows) in combination with adaptive learning model $A(y)$ (columns).

Read:

- In the late phase, the performance gap shrinks sharply. Table 11 shows average absolute prediction errors around 0.15 for the best learning models, versus 0.17 for equilibrium.
- Table 12 shows mean square prediction errors around 0.30–0.32 for the best models, including the equilibrium benchmark.
- Table 13 confirms the gap is tiny relative to the middle phase: the difference in average absolute prediction error is about 0.02, and the difference in mean square error is about 0.002.
- Figure 16 shows the learning and equilibrium price paths are now very close to one another, and both are close to the data.
- So the big substantive conclusion is phase-specific: learning matters a lot during convergence, but once the market settles down, equilibrium becomes a reasonable approximation.

TABLE 13—LATE PHASE: COMPARISON OF LEARNING MODEL ($F(0), A(\emptyset)$) AND COMPLETE INFORMATION NASH EQUILIBRIUM

	$CE_t^{(0,\emptyset)} - CE_t^*$	$(b_{j,t}^{(0,\emptyset)} - b_{j,t}^*)^2 - (b_{j,t}^* - b_{j,t})^2$
Constant	-0.023 (0.007)	-0.007 (0.014)
Time trend	-0.002 (0.001)	-0.002 (0.001)
Observations	29	29

Notes: In the first two regressions with $CE_t^{(0,\emptyset)} - CE_t^*$ as the dependent variable, the unit of observation is a month. In the last two regressions with $(b_{j,t}^{(0,\emptyset)} - b_{j,t}^*)^2 - (b_{j,t}^* - b_{j,t})^2$ as the dependent variable, the unit of observation is a BM unit-month. In the latter, observations are weighted by $\frac{q_{j,t}}{\sum_{j \in J} q_{j,t}}$ as described in Section VC and standard errors are clustered by month.

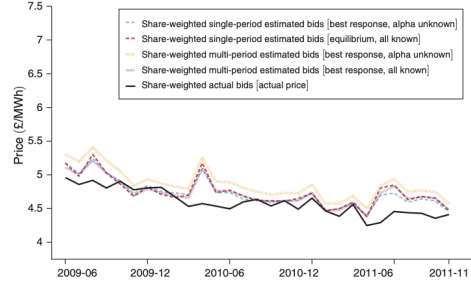


FIGURE 16. PREDICTED AND ACTUAL FR PRICE BY MONTH

Notes: Share-weighted average computed from bids predicted by fictitious play $F(0)$ with adaptive learning $A(\alpha)$ in both single and multi-period simulations, complete information Nash equilibrium without demand uncertainty $A(\emptyset)$, and actual bids (late phase).

5.9 Efficiency consequences and robustness

Read:

- The difference between actual and equilibrium average cost is tiny: £1.318/MWh in the data versus £1.3133/MWh in the equilibrium counterfactual.
- This means the main consequence of slow convergence is pricing and rents, not a large productive inefficiency.
- Robustness checks on fuel costs, BM repositioning, and collusion all leave the central message essentially unchanged.

6 Short summary

- **Method.** Estimate FR demand and cost primitives first, then use them to compare fictitious play plus adaptive learning against complete-information Nash equilibrium.
- **Identification.** The key empirical advantage is a newly deregulated market with fairly stable fundamentals for several years, so movements in bids can plausibly be attributed to learning.
- **Application.** In the UK FR market, firms spent years learning how rivals bid and how demand reacted to bids.
- **Main result.** Learning models outperform equilibrium in the transition phase, but once the market reaches a rest point, equilibrium fits about as well.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that a real market can take a long time to settle after deregulation: about three-and-a-half years, or 42 months, in this case.
- It also shows that **learning models can be disciplined empirically** using estimated demand and cost primitives, rather than just discussed abstractly.
- Finally, it establishes that weak-rationality methods can recover sensible cost estimates without imposing full equilibrium.

7.2 Economic intuition

- In a new market, firms do not know how aggressive rivals will be or how elastic demand is. They therefore use history to learn.
- Early on, firms experiment and behave heterogeneously. Once some firms undercut high bidders and take share, competitive pressure compresses bids.
- Because profits are not extremely sensitive to small bid mistakes, convergence is gradual rather than immediate.
- As history accumulates, recent rival behavior becomes informative enough that adaptive best responses guide firms toward the eventual rest point.

7.3 Takeaway

This paper is one of the clearest empirical demonstrations that the path to equilibrium matters. It is not enough to say that a newly designed market will end up near a Nash equilibrium; the transition can be long, strategically rich, and better described by learning rules than by equilibrium itself. For empirical IO and market design, the lesson is simple: after major institutional change, use equilibrium cautiously and take learning seriously.